INTRO TO ITWS — QUIZ 1 MASTER CRIB SHEET

QUICK LOOKUP INDEX

- CSS specificity → Inline > ID > Class > Element
- HTML vs XHTML → forgiving vs strict
- XML rules → close tags, root element
- Web Science → study of web + society
- Generative vs Agentic → answer vs action
- HTTP request cycle → DNS lookup → TCP handshake → HTTP request → HTTP response
- TCP/IP layers → Application / Transport / Internet / Link
- IPv4 vs IPv6 → 32-bit (~4.3B) vs 128-bit (~340 undecillion)
- Encapsulation → Wrapping data with headers at each layer
- CSS placement → External (best) vs Embedded vs Inline
- XML vs HTML → Data storage/transport vs Data display
- XHTML → HTML written using XML rules

Format: Plain-text + Markdown Design Goal: Maximum information density, fast lookup, exam-aligned explanations Chronological Order: By lecture / lab progression Keywords: Bolded All vocab defined

LECTURE 1 — INTRO TO ITWS & WEB SCIENCE

Core Concepts

ITWS (Information Technology & Web Science) studies how technology, people, and information systems interact at scale.

The course is not just coding — it includes networks, web architecture, social impact, economics, and governance.

The web is a socio-technical system: technical infrastructure + human behavior.

Key Vocabulary (Defined)

Web — A global system of interlinked hypertext documents accessed via the Internet using HTTP/HTTPS.

Internet — A global network of networks that communicate using the TCP/IP protocol suite.

Socio-technical system — A system involving both technical components and human/social components.

Scalability — The ability of a system to handle growth in users, data, or traffic without failure.

Emergent behavior — System-level outcomes that arise from many individual interactions (e.g., viral content).

Web Science

Web Science — An interdisciplinary field that studies:

How the web works

How it evolves

How it affects society

How society affects it

The Web Science Method

Observe a web-based phenomenon

Form a hypothesis

Collect data

Analyze results

Intervene or propose solutions

Evaluate outcomes

Exam angle: used to analyze social problems, misinformation, network effects.

How This Appears on the Quiz

Q: What is Web Science and why does it matter?

Answer:

Q: How is ITWS different from a programming-only course?

Answer:

Q: Explain how the web is a socio-technical system.

Answer:

LECTURE 2 — NETWORKS, PROTOCOLS, TCP/IP

Core Concepts

Communication over the internet happens in layers.

Each layer has a specific responsibility.

Data is wrapped as it moves down the stack and unwrapped as it moves up.

What is a Network?

A network transfers information between devices.

A switch connects devices within a network.

A router connects different networks.

Key Vocabulary (Defined)

Protocol — A formal set of rules governing how data is formatted, transmitted, and received.

Packet — A formatted unit of data transmitted over a network.

Encapsulation — Wrapping data with protocol-specific headers at each layer.

Latency — Delay in data transmission.

Bandwidth — Maximum data transfer capacity.

TCP/IP Model (4 Layers)

```
Application → Transport → Internet → Link
HTTP        → TCP/UDP   → IP        → Ethernet/WiFi
```

What this does: Shows a protocol or layered model used to move data reliably across networks.

TCP Three-Way Handshake

SYN — Client requests connection

SYN-ACK — Server acknowledges and agrees

ACK — Client confirms

PURPOSE: Ensures both sides are ready before data transfer.

HTTP Request Cycle

DNS lookup (usually UDP)

TCP handshake

```
HTTP GET request
HTTP response
```

What this does: Demonstrates the correct syntax and structure for the concept in this section.

Connection maintained or closed

Key Vocabulary (Networking)

DNS — Domain Name System; maps domain names to IP addresses.

IP Address — Numerical identifier for a device on a network.

IPv4 — 32-bit addressing (~4.3 billion addresses).

IPv6 — 128-bit addressing (~340 undecillion addresses).

MAC Address — Hardware-level identifier for network interfaces.

How This Appears on the Quiz

Q: How does your computer know where to go when you type a URL?

Answer:

Q: Why is the TCP three-way handshake required?

Answer:

Q: Explain the role of DNS in web browsing.

Answer:

LECTURE 3 — HTML, XHTML, HTML5

Core Concepts

HTML is semantic markup, not just presentation.

Tags describe meaning, not appearance.

Browsers interpret structure; CSS controls styling.

HTML Versions

HTML 4.01 — Older, presentation-heavy

XHTML 1.0 — HTML rewritten as strict XML

HTML5 — Modern, semantic, extensible, “living standard”

Key Vocabulary (Defined)

Markup language — A language that annotates text with tags to describe structure and meaning.

Semantic HTML — HTML that conveys meaning (e.g., <header>, <nav>, <article>).

DOCTYPE — Tells the browser what type/version of document to expect.

Metadata — Data that describes data.

Basic HTML5 Structure (Annotated)

```
<!-- PURPOSE: Declares the document type as HTML5 -->
<!DOCTYPE html>
<!-- PURPOSE: Root element of the HTML document -->
<html>
<!-- PURPOSE: Contains metadata, styles, scripts, and title -->
<head>
<title>Page Title</title> <!-- PURPOSE: Appears in browser tab -->
</head>
<!-- PURPOSE: Contains all visible page content -->
<body>
<p>Hello World</p> <!-- PURPOSE: Displays a paragraph -->
</body>
</html>
```

What this does: Provides structured markup so browsers (and tools like search engines or screen readers) can interpret the page correctly.

Block vs Inline Elements

Block elements — Take full width, create line breaks (<p>, <div>, <h1>)

Inline elements — Flow within text (<a>, <em>, <strong>)

How This Appears on the Quiz

Q: What is the difference between HTML and XHTML?

Answer:

Q: What is the fundamental advantage of HTML5?

Answer:

Q: Why is semantic markup important?

Answer:

LECTURE 4 — CSS & SELECTORS

Core Concepts

CSS controls presentation, not content.

Multiple rules may apply → cascade & specificity decide winner.

Three Ways to Use CSS

External stylesheet (BEST)

Embedded <style> in <head>

Inline style="" (avoid)

CSS Rule Structure

```
/* PURPOSE: Selects paragraph elements */
p {
```

What this does: Applies presentation rules to selected elements; specificity and order determine which style is used.

color: red; /* PURPOSE: Sets text color */

}

Selectors & Specificity (High Exam Value)

Specificity order (highest → lowest):

Inline styles

ID selectors (#id)

Class selectors (.class)

Element selectors (p)

Later rules override earlier ones if specificity is equal.

Example Quiz Scenario (Explained)

```
<div id="mainBlock" class="header blockTitle">
<p>Main Title</p>
</div>
p { color: red; }                  /* element selector */
#blockTitle { color: blue; }       /* ID selector (does NOT match) */
```

What this does: Demonstrates the correct syntax and structure for the concept in this section.

p#mainBlock.header.blockTitle {

color: green;

}

Result: Red, because the ID selector does not apply and the compound selector is invalid.

CSS Box Model

Content

Padding

Border

Margin

How This Appears on the Quiz

Q: Which CSS rule wins and why?

Answer:

Q: Explain CSS specificity.

Answer:

Q: Why are external stylesheets preferred?

Answer:

LECTURE 5 — XML & STRUCTURED DATA

Core Concepts

XML describes data, not presentation.

Tags are user-defined.

XML is strict and rule-based.

Key Vocabulary (Defined)

XML — Extensible Markup Language.

Well-formed — Follows XML syntax rules.

DTD — Document Type Definition; defines valid structure.

Schema — XML-based alternative to DTD.

Entity — Encoded reserved character (<, &).

XML Rules (Exam Favorites)

Tags must close

Case-sensitive

Attributes must be quoted

Single root element

Proper nesting

XML Example (Annotated)

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- PURPOSE: XML declaration -->
<book>
<title>Example</title> <!-- PURPOSE: Data element -->
</book>
```

What this does: Represents structured data using strict, nested tags for reliable machine parsing.

XML vs HTML

How This Appears on the Quiz

Q: Why use XML instead of HTML?

Answer:

Q: What does it mean for XML to be well-formed?

Answer:

Q: How are XML and XHTML related?

Answer:

LECTURE 6 — AI: GENERATIVE VS AGENTIC

Core Concepts

AI tools differ in control vs autonomy.

Understanding the difference matters professionally and ethically.

Key Vocabulary (Defined)

Generative AI — Produces outputs in response to prompts (code, text).

Agentic AI — Executes tasks autonomously toward a goal.

Prompt engineering — Crafting effective inputs to guide AI output.

Comparison

CLEAR Prompt Method

Concise

Logical

Explicit

Adaptive

Reflective

How This Appears on the Quiz

Q: Compare generative and agentic AI.

Answer:

Q: What are the risks of over-reliance on agentic AI?

Answer:

Q: Why must developers still understand fundamentals when using AI?

Answer:

LABS — WEBSITE & VM DEPLOYMENT

Core Concepts

Permissions matter

Servers run as users

Authentication controls access

Key Commands (Annotated)

```
sudo chown -R www-data:www-data /var/www/html
# PURPOSE: Gives Apache ownership so it can read/write files
sudo usermod -a -G www-data yourRCSid
# PURPOSE: Adds user to web server group for permissions
sudo nano /var/www/html/iit/.htaccess
# PURPOSE: Configures directory-level authentication
```

What this does: Demonstrates the correct syntax and structure for the concept in this section.

How This Appears on the Quiz

“Describe steps to modify site access.”

“Explain why permissions are required.”

“What does Apache need to serve pages?”

CASE DISCUSSION — KUEHCHIC DESSERTS

Key Concepts

Stakeholders — Lim, Tan, Kumar, customers, suppliers.

Triple Constraint — Scope, Time, Cost.

Critical Path — Longest sequence of dependent tasks.

Free Float — Delay allowed without affecting successors.

Total Float — Delay allowed without affecting project end.

How This Appears on the Quiz

Q: Identify the stakeholders and their influence.

Answer:

Q: Which project constraint is most important and why?

Answer:

Q: What are the major project risks?

Answer:

QUICK LOOKUP INDEX (PANIC SECTION)

TCP handshake → SYN / SYN-ACK / ACK

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Web Science → study of web + society

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